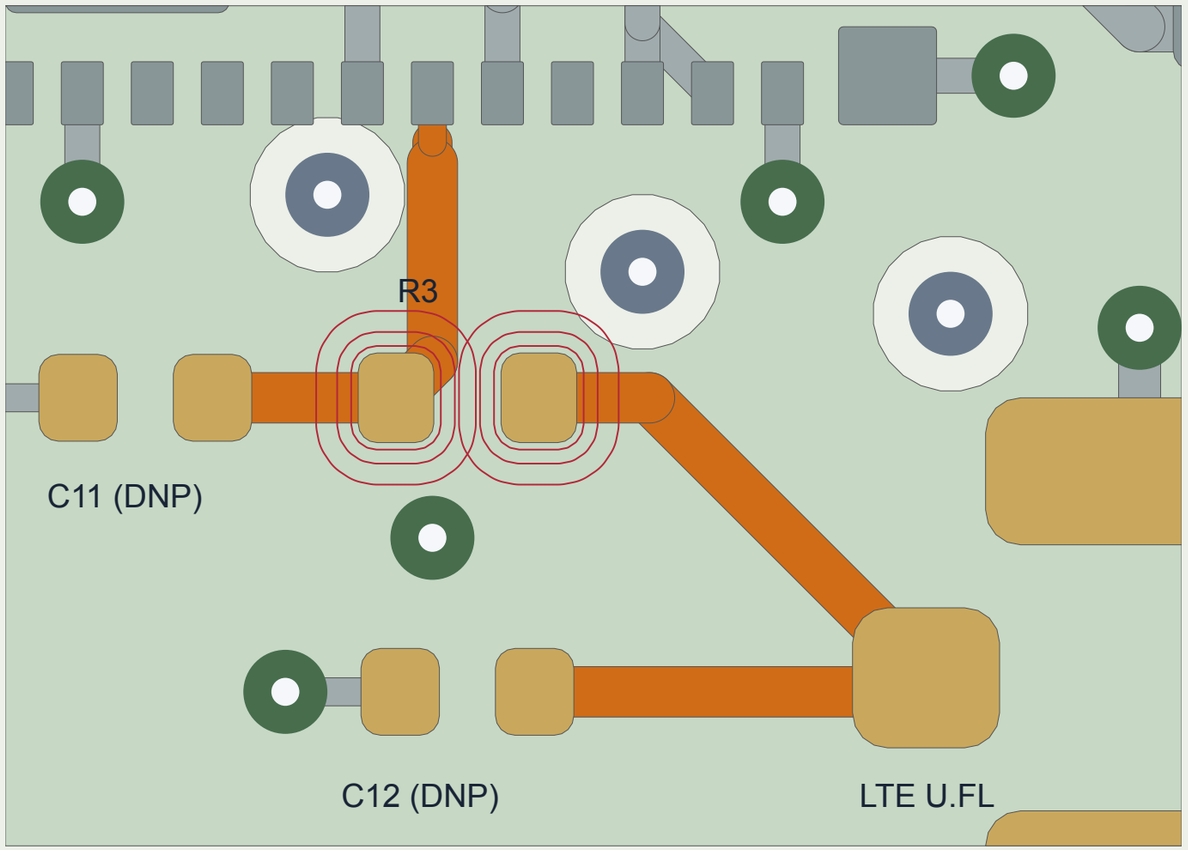
Collar tracker — R3 and LTE tuning junctions

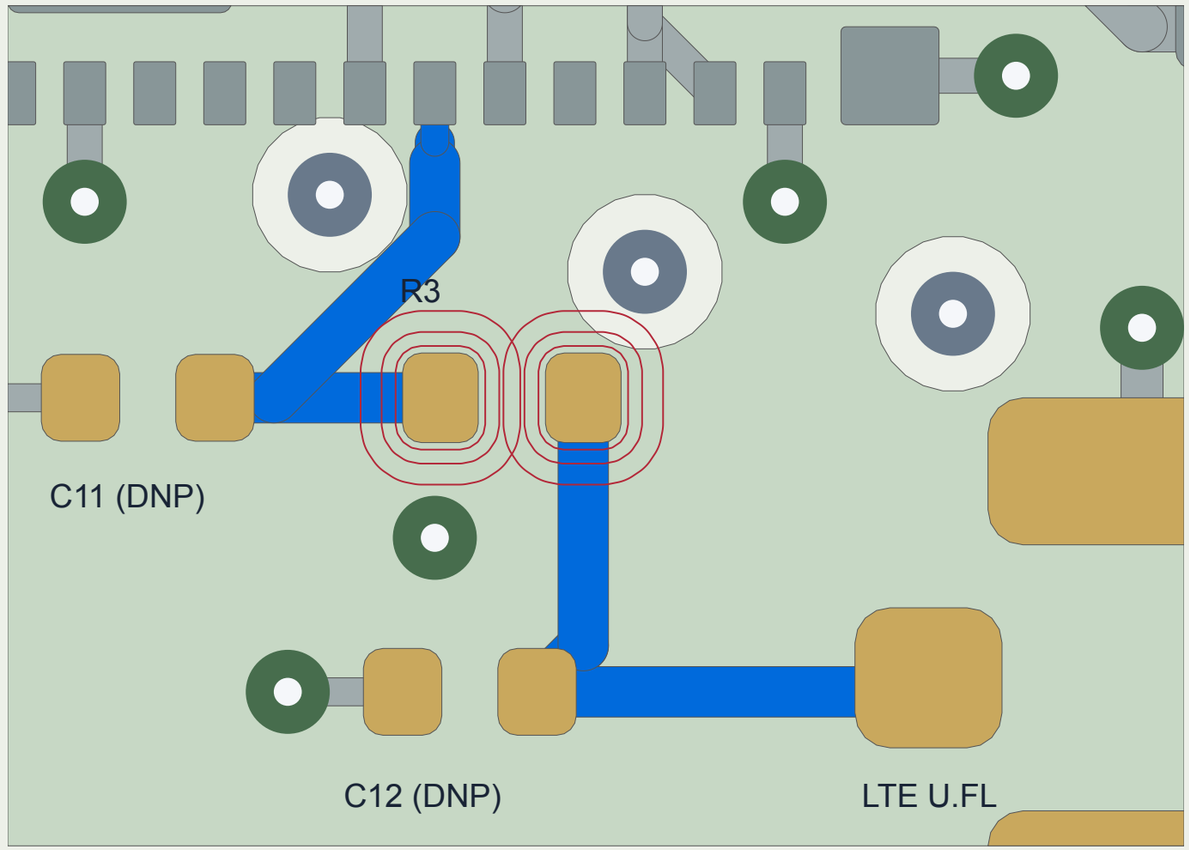
r05j: one 0.36 mm entry per R3 land; the resistor moves 0.30 mm. Native DRC/ERC/parity: zero.

Before — r05i



R3 ratios: 2.288 / 2.413 / 2.090

After — r05j



R3 ratios: 1.004 / 1.003 / 1.001

Red contours are 0.05 / 0.15 / 0.30 mm outside each land. Ratios compare unique trace-copper intersections.

C11 branch: 1.31 → 0.42 mm centerline. C12 lies on the main path, eliminating its previous 2.84 mm branch.

Tradeoff: combined LTE feed centerline grows from 5.89 to 8.91 mm, excluding the resistor body. RF loss is unmeasured.

All vias, other component placements, inner copper, continuous GND planes, module escapes and stack stay unchanged.

1,742 native reference probes agree; all RF trace copper projects onto In1.Cu GND. This is not an EM simulation.

69 populated envelopes pass the modeled thickness range. Assembly, power, antenna/body and IP68 remain unqualified.

PCB SHA256 51fc05ccc66009995daa8c5d5466aa983436e00e7ff338f0ddfb0927c92b4580